Soham Mukherjee

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Professional Summary

Lead Software Engineer and industry AI/ML researcher (PhD, Purdue University) developing Physical AI models for Electronic Design Automation. Builds physics-grounded neural surrogates and graph-learning systems for circuit simulation, spanning model architecture, synthetic-data generation, large-scale training, and out-of-distribution evaluation. Research foundation in geometric and topological deep learning.

Work Experience

Lead Software Engineer, Cadence Design Systems, San Jose, CA Feb 2024 - Present

- Developed **Spectre Conductor**, an agentic knowledge system for navigating a large circuit-simulator codebase using grounded retrieval, API/MCP integration, and rigorous evaluation, reducing token usage by **up to 90%**.
- Built CNN and GNN surrogate models for physics-grounded field prediction of circuit voltage solutions, achieving **0.35% global relative L2 error** on out-of-distribution test data, and characterized neural scaling laws across task complexity using automated data generation.
- Developed graph-learning models to predict circuit-simulation runtime and memory directly from attributed circuit graphs, achieving **up to 0.99 R^2** for peak-memory prediction while benchmarking GNN, contrastive, multimodal, and state-space architectures.
- Built a graph data-augmentation pipeline that synthesized **15,050 labeled circuits from 717 originals**, expanding scarce post-layout data by **21x** for simulator runtime and memory prediction.
- Developed triplet-loss pretraining for graph embeddings and log-processing pipelines spanning **5,592 device types** for sparse-target learning.
- Scaled model training to graphs with up to **1M nodes** across GPUs using containerized workflows, MLflow experiment tracking, and versioned datasets.

AI Research Intern, IBM T.J. Watson, Yorktown Heights, NY May 2022 - Aug 2022

- Developed constrained graph-generation methods that led to a **published U.S. patent application**.

AI Engineering Intern, Physna Inc., Columbus, OH May 2021 - Aug 2021

- Deployed **CNNs for 3D CAD prediction** from 2D images.
- Automated **point-cloud segmentation and registration** for efficient and accurate inspection of scanned machine parts.

Education

PhD Computer Science | Purdue University *(Awarded Scholarship)* Aug 2020 - Dec 2023

MS Computer Science | Ohio State University Aug 2017 - Jul 2020

BE Electronics & Telecommunication | Jadavpur University *(Gold Medalist)* Aug 2013 - Aug 2017

Skills

Languages: Python, C/C++, Java
Frameworks & Systems: PyTorch, PyTorch-Geometric, TensorFlow, MLflow, Distributed GPU Training
Physical AI & Model Development: Neural Surrogates, GNNs, Neural Operators, Physics-Grounded Learning, Multimodal and Contrastive Learning, State-Space Models, Synthetic-Data Generation, OOD Evaluation
Research Areas: Scientific ML, Circuit Modeling, Topological Deep Learning, Computer Vision, NLP, and Agentic AI

Selected Patents & Publications

1. Mukherjee, Soham, Karthikeyan Natesan Ramamurthy, and Payel Das. 'Generative Modeling with Topological Control'. U.S. Patent Application Publication US 2026/0187916 A1, 2026.
2. Mukherjee, Soham, Shreyas N. Samaga, et al. 'D-GRIL: End-to-End Topological Learning with 2-Parameter Persistence'. Proceedings of the 42nd International Symposium on Computational Geometry (SoCG 2026), 2026.
3. Zhang, Simon, et al. 'GEFL: Extended Filtration Learning for Graph Classification'. Learning on Graphs Conference, PMLR, 2022.
4. Hajij, Mustafa, et al. 'Topological Deep Learning: Going Beyond Graph Data'. arXiv Preprint arXiv:2206.00606, 2023.
5. Xin, Cheng, et al. 'GRIL: A 2-Parameter Persistence Based Vectorization for Machine Learning'. Proceedings of the 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML), vol. 221, 2023.

Full publication list is available on Google Scholar.